Assignment 3 – Docker Compose Multi-Container Application

GitHub Repository

Repository Name: devops-assignment-3-docker-compose

Implementation Steps

1. Created project folder for Assignment 3 in WSL.
2. Created docker-compose.yml file for multi-container setup.
3. Added backend, frontend, MySQL and Redis services.
4. Built the backend image using Dockerfile.
5. Ran docker compose up -d command.
6. Verified all containers started successfully.
7. Tested Redis connection using redis-cli ping command and got PONG.

Configuration Files Used

1. docker-compose.yml – service definitions 2. Dockerfile – backend image build 3. requirements.txt – Python dependencies 4. app configuration files

Troubleshooting Guide

• Warning about version attribute: Docker Compose showed obsolete warning but containers still worked. • Build issue: checked Dockerfile path. • Container not starting: checked docker compose logs.

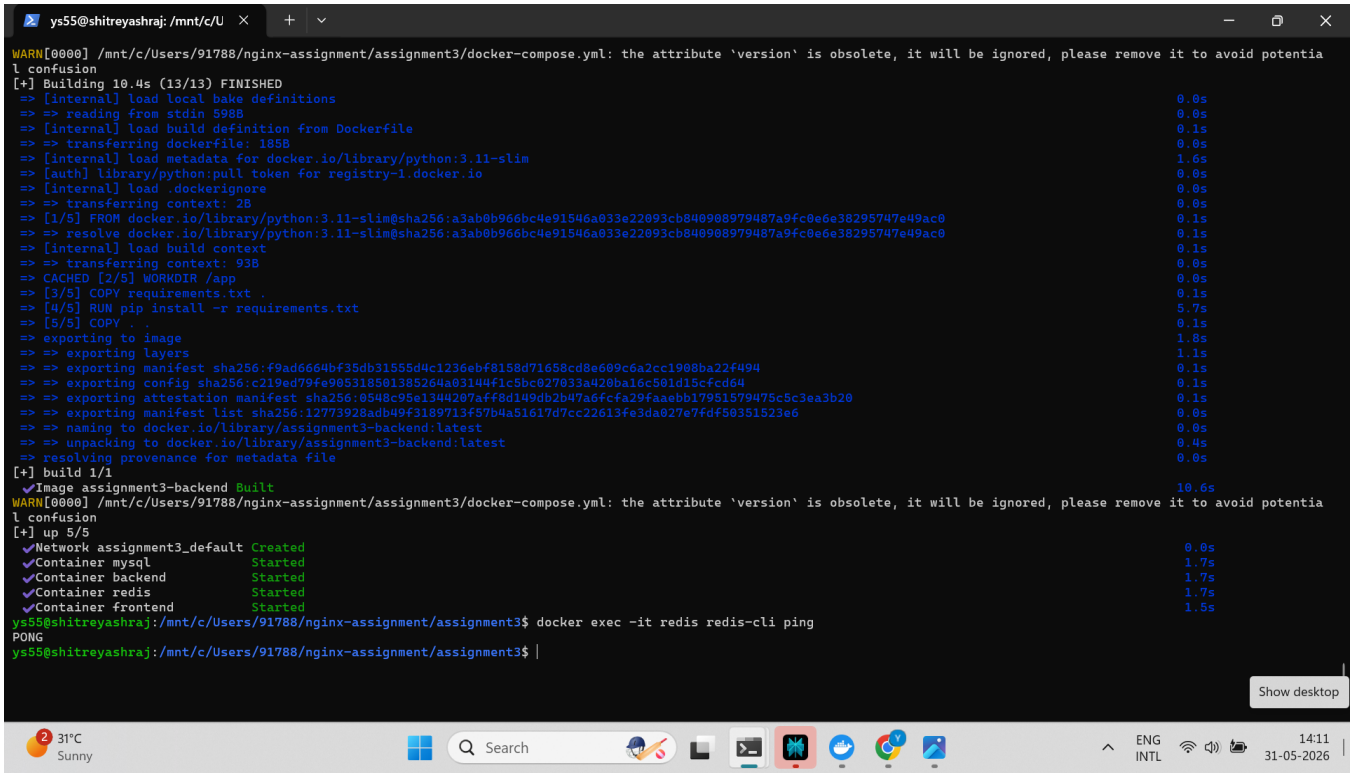
Rollback / Recovery

Stopped all services using docker compose down. Updated config if needed and started again using docker compose up -d.

Key Decisions

Docker Compose was used because it helps manage multiple services together. Redis and MySQL were added for service communication and testing. This setup is useful for real DevOps deployment practice.

Screenshot


ys55@shitreyashraj: /mnt/c/U
WARN[0000] /mnt/c/Users/91788/nginx-assignment/assignment3/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potentia
l confusion
[+] Building 10.4s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 598B 0.0s
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 185B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 1.6s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0 0.1s
=> => resolve docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 93B 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> [3/5] COPY requirements.txt . 0.1s
=> [4/5] RUN pip install -r requirements.txt 5.7s
=> [5/5] COPY . . 0.1s
=> exporting to image 1.8s
=> => exporting layers 1.1s
=> => exporting manifest sha256:f9ad6664bf35db31555d4c1236ebf8158d71658cd8e609c6a2cc1908ba22f494 0.1s
=> => exporting config sha256:c219ed79fe905318501385264a03144f1c5bc027033a420ba16c501d15cfcd64 0.1s
=> => exporting attestation manifest sha256:0548c95e1344207aff8d149db2b47a6fcfa29faaebb17951579475c5c3ea3b20 0.1s
=> => exporting manifest list sha256:12773928adb49f3189713f57b4a51617d7cc22613fe3da027e7fdf50351523e6 0.0s
=> => naming to docker.io/library/assignment3-backend:latest 0.0s
=> => unpacking to docker.io/library/assignment3-backend:latest 0.4s
=> resolving provenance for metadata file 0.0s
[+] build 1/1
✔Image assignment3-backend Built 10.6s
WARN[0000] /mnt/c/Users/91788/nginx-assignment/assignment3/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potentia
l confusion
[+] up 5/5
✔Network assignment3_default Created 0.0s
✔Container mysql Started 1.7s
✔Container backend Started 1.7s
✔Container redis Started 1.7s
✔Container frontend Started 1.5s
ys55@shitreyashraj:/mnt/c/Users/91788/nginx-assignment/assignment3$ docker exec -it redis redis-cli ping
PONG
ys55@shitreyashraj:/mnt/c/Users/91788/nginx-assignment/assignment3$
Show desktop
31°C
Sunny
Search
ENG
INTL
14:11
31-05-2026